

Signature:

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

End-of-Module Exam: 2027 Spring Semester

Zürcher Hochschule

für Angewandte Wissenschaften

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Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX08-WIN.XX / Data Science Introduction
Exam date / time:	25.08.2027 / 14:00 – 15:00
Duration:	60 Minutes
Total number of pages:	2
Permissible resources:	Closed book
Calculator:	no calculator
Dictionary:	Yes

General information:

- Use a black or blue pen (invisible ink is not allowed) to write your answers.
- All options of a question (i.e. A–D) can be ticked.
- ONLY tick ONE option per question in the table.
- Scoring: each correctly answered question = 1 point.
- Write your answers only on the answer sheet provided; rough work on question pages is not scored.
- Switch off your mobile phone and put it in your bag for the duration of the examination.
- If you want to write a draft, use the back of the question pages. Only the answer sheet counts.
- Please remain seated until all exams have been collected.

Marked by:	Total points:	Grade:

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Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1	100%		
		<i>(Grade on cover sheet)</i>	
Module grade (Evento)			⇐ Rounded ⇐

Marking Table

No.	Task	Points	Corrector's, initials
1	MC written Exam	30 Points	hitz
	Total	30 Points	

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Answer sheet

ONLY the answer sheet will be scored. All options of a question (i.e. A–D) can be ticked.

ONLY tick ONE option per question in this table.

Scoring: Each correctly answered question is worth **1 point**.

Example: For question 0, C should be ticked, then it would look like this.

Question	Answer	Question	Answer
1.	☐ A ☐ B ☐ C ☐ D	16.	☐ A ☐ B ☐ C ☐ D
2.	☐ A ☐ B ☐ C ☐ D	17.	☐ A ☐ B ☐ C ☐ D
3.	☐ A ☐ B ☐ C ☐ D	18.	☐ A ☐ B ☐ C ☐ D
4.	☐ A ☐ B ☐ C ☐ D	19.	☐ A ☐ B ☐ C ☐ D
5.	☐ A ☐ B ☐ C ☐ D	20.	☐ A ☐ B ☐ C ☐ D
6.	☐ A ☐ B ☐ C ☐ D	21.	☐ A ☐ B ☐ C ☐ D
7.	☐ A ☐ B ☐ C ☐ D	22.	☐ A ☐ B ☐ C ☐ D
8.	☐ A ☐ B ☐ C ☐ D	23.	☐ A ☐ B ☐ C ☐ D
9.	☐ A ☐ B ☐ C ☐ D	24.	☐ A ☐ B ☐ C ☐ D
10.	☐ A ☐ B ☐ C ☐ D	25.	☐ A ☐ B ☐ C ☐ D
11.	☐ A ☐ B ☐ C ☐ D	26.	☐ A ☐ B ☐ C ☐ D
12.	☐ A ☐ B ☐ C ☐ D	27.	☐ A ☐ B ☐ C ☐ D
13.	☐ A ☐ B ☐ C ☐ D	28.	☐ A ☐ B ☐ C ☐ D
14.	☐ A ☐ B ☐ C ☐ D	29.	☐ A ☐ B ☐ C ☐ D
15.	☐ A ☐ B ☐ C ☐ D	30.	☐ A ☐ B ☐ C ☐ D

MC Data Science Introduction (30 Points)

1. Which statement comparing the VSM and System Dynamics is NOT correct? (1)
 - A. Both can make use of balancing and reinforcing feedback loops.
 - B. System Dynamics focuses on dynamic behaviour and uses causal loop diagrams.
 - C. The VSM focuses on steering, viability and organisational recursion.
 - D. The VSM and System Dynamics both reject the concept of feedback.
2. In the light of Ashby's Law, what does 'adaptation' mean? (1)
 - A. Collecting more data without changing capacity.
 - B. Reducing the system to a single fixed state.
 - C. Increasing one's own variety (or attenuating the environment's) so the system matches environmental variety.
 - D. Ignoring environmental change.
3. The DIKIW hierarchy that frames Data Science runs in which order? (1)
 - A. Data -> Knowledge -> Information -> Wisdom -> Intelligence.
 - B. Information -> Data -> Wisdom -> Knowledge -> Intelligence.
 - C. Data -> Information -> Knowledge -> Intelligence -> Wisdom.
 - D. Wisdom -> Intelligence -> Knowledge -> Information -> Data.
4. Which single statement about real-time adaptive models is correct? (1)
 - A. Concept drift is irrelevant to adaptive models.
 - B. Truly adaptive models require no feedback at all.
 - C. Model-free reinforcement learning learns purely from a reward history.
 - D. Adaptive models are only static once-trained reports.
5. Which statement about overfitting is NOT correct? (1)
 - A. It reflects high variance.
 - B. Cross-validation helps detect it.
 - C. It is best fixed by making the model even more complex.
 - D. It generalizes poorly to new data.
6. Which statement about visualization pitfalls is NOT correct? (1)
 - A. They can imply false completeness or logic.
 - B. They are immune to the designer's conscious or unconscious bias.
 - C. They require visual literacy to interpret correctly.
 - D. Visualizations can manipulate emotions rather than inform objectively.

7. Which of the following is a hallmark of complex CPS data? (1)
- A. Small, perfectly structured, independent samples.
 - B. Large-scale, high-frequency, heterogeneous and non-IID data.
 - C. A single clean spreadsheet.
 - D. Static data with no time dependency.
8. Compared with SMOTE, class weighting... (1)
- A. duplicates minority-class rows.
 - B. leaves the data unchanged and adjusts the loss instead.
 - C. creates synthetic samples by interpolation.
 - D. deletes majority-class rows.
9. Which single statement about target-oriented action is correct? (1)
- A. The available data should dictate the goal.
 - B. Goals should be measurable and stakeholders aligned on the same target.
 - C. 'Playing with data' without a purpose is the recommended start.
 - D. Analysis is the end goal; action is unnecessary.
10. Which statement about explainability is NOT correct? (1)
- A. Opaque black-box models are ideal for high-stakes use.
 - B. XAI explains the 'why' behind decisions.
 - C. It is key to trust and transparency.
 - D. Explanations should adapt to the audience.
11. Which statement about CPS control is NOT correct? (1)
- A. Control and regulation rely on feedback and information flow.
 - B. A controller may have far less variety than the system and still control it.
 - C. CPS control must work across multiple hierarchical levels.
 - D. High-quality models enable effective control of complex systems.
12. The Viable System Model (VSM) is best described as... (1)
- A. a star schema for data warehousing.
 - B. a one-off linear plan.
 - C. a recursive management structure that ensures viability via anti-oscillatory feedback.
 - D. a confusion matrix.
13. Which of the following is a typical class-imbalance scenario? (1)
- A. Predicting house prices from size.
 - B. Normalizing numeric features.
 - C. Clustering customers by behaviour.
 - D. Fraud detection where positive cases are about 1% of the data.

14. Which single statement about acting under uncertainty is correct? (1)
- A. Under uncertainty, analysis is fundamentally impossible.
 - B. Scenario analysis helps cope with uncertain outcomes.
 - C. Uncertainty is just another word for volatility.
 - D. Uncertainty means a total absence of data.
15. Which statement about unsupervised learning is NOT correct? (1)
- A. It is harder to evaluate than supervised learning.
 - B. It works on unlabeled data.
 - C. Clustering is a common application.
 - D. It requires labels for every example.
16. Which statement about requisite variety is NOT correct? (1)
- A. The law guarantees control as long as enough data is collected.
 - B. Variety is a measure of complexity (the number of distinguishable states).
 - C. 'Fatal undercomplexity' is having too little variety to cope.
 - D. If environmental variety exceeds capacity, the environment dominates the system.
17. System Dynamics studies... (1)
- A. how systems behave over time via stocks, flows, feedback and delays.
 - B. labeled classification problems.
 - C. static snapshots only.
 - D. database table schemas.
18. In a smart factory, forecasting a bearing failure from vibration data is an example of... (1)
- A. descriptive reporting.
 - B. predictive maintenance.
 - C. unsupervised clustering.
 - D. manual control.
19. Which single statement about model evaluation is correct? (1)
- A. The F1 score balances precision and recall and is useful under imbalance.
 - B. Accuracy is always a sufficient metric.
 - C. A confusion matrix applies only to regression.
 - D. An ROC curve closer to the diagonal indicates a stronger model.
20. Which statement about unsupervised algorithms is NOT correct? (1)
- A. For K-means, k is chosen beforehand.
 - B. K-means uses centroids as cluster centres.
 - C. K-medoids is more robust to outliers than K-means.
 - D. Fuzzy C-means forces each point into exactly one cluster.

21. Which statement about the target-centric approach/strategy is NOT correct? (1)
- A. It is identical to a generic, purely system-centered approach.
 - B. It improves accountability and performance tracking.
 - C. Resources are allocated according to clear, strategy-derived priorities.
 - D. BI systems are built to reflect and support enterprise goals.
22. Which best distinguishes clustering from network analysis? (1)
- A. Clustering needs labels and network analysis is supervised.
 - B. Both require fully labeled data.
 - C. They are identical techniques.
 - D. Clustering groups similar items, while network analysis studies the connections between items.
23. Without metadata management, a data lake tends to become... (1)
- A. a data swamp.
 - B. a star schema.
 - C. a Golden Record.
 - D. a confusion matrix.
24. Which single statement about EDA is correct? (1)
- A. EDA is unnecessary whenever the dataset is large.
 - B. EDA is the final deployment phase.
 - C. EDA examines distributions, missing values and outliers to guide later steps.
 - D. EDA steps may safely use test-set information during training.
25. Which statement about Ashby's Law is a misinterpretation (i.e., NOT correct)? (1)
- A. Variety measures the number of distinguishable states.
 - B. You can attenuate environmental variety instead of amplifying your own.
 - C. The controller should match the system's variety.
 - D. Variety is simply the volume of data you hold.
26. Which statement about the VUCA environment is NOT correct? (1)
- A. Preparation is futile under VUCA conditions.
 - B. Each dimension requires its own tailored response.
 - C. The term was first introduced by the US military.
 - D. System dynamics helps navigate VUCA conditions.
27. Which of the following is a lazy learner? (1)
- A. Logistic regression
 - B. Neural network
 - C. Decision tree
 - D. K-nearest neighbors (KNN)

28. A hiring model trained on biased historical decisions keeps perpetuating discrimination. This is mainly... (1)
- A. historic plus algorithmic bias.
 - B. random sampling noise.
 - C. no bias at all.
 - D. overfitting only.
29. Which single statement about early warning in viable/CPS systems is correct? (1)
- A. Early warning is impossible in cyber-physical systems.
 - B. Feedback monitors the system's state so deviations are caught while correction is still possible.
 - C. Early warning requires removing all feedback loops.
 - D. A system without feedback warns earliest.
30. Which statement about interdisciplinarity is NOT correct? (1)
- A. Social science contributes ethics and fairness perspectives.
 - B. Data Science draws on multiple disciplines.
 - C. Data Science is purely a branch of software engineering.
 - D. Domain knowledge helps prevent 'solitary analysis'.